Lahore University of Management Sciences
Syed Babar Ali School of Science and Engineering
AI-500 and AI-501
Fall 2024
Mid Exam – Part II

Weightage: 60%

Name: _____

Campus ID: _____

Note:
There are ten questions worth 70 points. We require you to attempt questions with a cumulative worth of 60 points. Do not over-attempt; we will not grade the one with the highest marks.

Problem 1. (**6 pts**) Consider a full HD image represented as a matrix Z of size 1920×1080 (rows $\times$ columns). Assume that the matrix Z has only 120 non-zero singular values. Using a **low-rank approximation** with rank $k = 120$, determine the **storage savings** achieved in terms of the **number of pixels** (or matrix entries) that need to be stored.

1. What is the total number of entries in the original matrix Z?
2. How many entries need to be stored for the low-rank approximation of rank $k = 120$?
3. Calculate the percentage reduction in storage enabled by this low-rank approximation.

Solution:

1. Total entries in the original matrix Z:

Since the matrix Z is of size 1920×1080, the total number of entries is $1920 \times 1080 = 2,073,600$ entries

2. Entries required for the low-rank approximation of rank $k = 120$:

The low-rank approximation of rank k requires storing the following:

- U: A 1920×120 matrix, requiring 1920×120 entries.
- Σ: A 120×120 diagonal matrix, requiring 120 entries.
- $V^\top$: A 120×1080 matrix, requiring 120×1080 entries.

The total number of entries required for the low-rank approximation is:

$$1920 \times 120 + 120 + 120 \times 1080 = 360,120.$$

3. Percentage reduction in storage:

The storage savings can be computed as:

$$\text{Savings} = \left(1 - \frac{360,120}{2,073,600}\right) \times 100 = 82.64\%.$$

Problem 2. (**4 pts**) Suppose A is a 5×10 matrix, B is a 20×10 matrix, and C is a 10×10 matrix. Determine whether each of the following expressions make sense. If the expression makes sense, give its dimensions.

(a) $A^T A + C$.

(b) BC^3.

Solution:

(a) $A^T A + C$: A^T is a 10×5 matrix, so $A^T A$ is:

$$A^T A : (10 \times 5) \cdot (5 \times 10) = 10 \times 10.$$

Since C is also a 10×10 matrix, the sum $A^T A + C$ makes sense and the result is:

$$\text{Dimension: } 10 \times 10.$$

(b) BC^3: First, $C^3 = C \cdot C \cdot C$ is a product of three 10×10 matrices, which results in:

$$C^3 : (10 \times 10) \cdot (10 \times 10) \cdot (10 \times 10) = 10 \times 10.$$

Now, multiply B (a 20×10 matrix) with C^3 (a 10×10 matrix):

$$BC^3 : (20 \times 10) \cdot (10 \times 10) = 20 \times 10.$$

The expression makes sense, and its dimension is:

$$\text{Dimension: } 20 \times 10.$$

Problem 3. (**6 pts**) This problem is related to orthogonal matrices.

(a) Prove that the inverse of an orthogonal matrix is equal to its transpose.

(b) Show that the orthogonal matrix preserves the norm and angles between the vectors.

Solution:

(a) Let Q be an $n \times n$ orthogonal matrix. By definition:

$$Q^\top Q = I.$$

Multiplying both sides on the right by Q^{-1}:

$$Q^\top = Q^{-1}.$$

Therefore, the inverse of Q is equal to its transpose.

(b) For any vector $\mathbf{x} \in \mathbb{R}^n$:

$$\|Q\mathbf{x}\| = \|\mathbf{x}\|,$$

because:

$$\|Q\mathbf{x}\|^2 = (Q\mathbf{x})^\top (Q\mathbf{x}) = \mathbf{x}^\top Q^\top Q\mathbf{x} = \mathbf{x}^\top \mathbf{x}.$$

Thus, orthogonal matrices preserve vector norms and angles between vectors.

Problem 4. (**8 pts**)

Apply the Gram-Schmidt algorithm to the following set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ to obtain an orthonormal set of vectors. Also determine the dimension of the subspace spanned by $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 2.5 \\ -2.5 \\ 5 \end{bmatrix}.$$

Solution: Step 1: Compute $\mathbf{q}_1$.

Normalize $\mathbf{v}_1$:

$$\mathbf{q}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}.$$

Step 2: Compute $\mathbf{q}_2$.

Compute the projection of $\mathbf{v}_2$ onto $\mathbf{q}_1$:

$$\text{proj}_{\mathbf{q}_1} \mathbf{v}_2 = (\mathbf{v}_2 \cdot \mathbf{q}_1)\mathbf{q}_1.$$

Calculate $\mathbf{v}_2 \cdot \mathbf{q}_1$:

$$\mathbf{v}_2 \cdot \mathbf{q}_1 = \frac{1}{\sqrt{6}}(0 \times 1 + 1 \times (-1) + 1 \times 2) = \frac{1}{\sqrt{6}}(1).$$

Compute the projection:

$$\text{proj}_{\mathbf{q}_1} \mathbf{v}_2 = \left(\frac{1}{\sqrt{6}}\right) \mathbf{u}_1 = \frac{1}{6} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}.$$

Find $\tilde{\mathbf{q}}_2$:

$$\tilde{\mathbf{q}}_2 = \mathbf{v}_2 - \text{proj}_{\mathbf{q}_1} \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{6} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{6} \\ \frac{7}{6} \\ \frac{2}{3} \end{bmatrix}.$$

Normalize $\tilde{\mathbf{q}}_2$:

$$\mathbf{q}_2 = \frac{\tilde{\mathbf{q}}_2}{\|\tilde{\mathbf{q}}_2\|} = \frac{1}{\sqrt{66}} \begin{bmatrix} -1 \\ 7 \\ 4 \end{bmatrix}.$$

Step 3: Compute $\mathbf{q}_3$.

Compute projections onto $\mathbf{q}_1$ and $\mathbf{q}_2$:

$$\text{proj}_{\mathbf{q}_1} \mathbf{v}_3 = (\mathbf{v}_3 \cdot \mathbf{q}_1)\mathbf{q}_1.$$

Calculate $\mathbf{v}_3 \cdot \mathbf{q}_1$:

$$\mathbf{v}_3 \cdot \mathbf{q}_1 = \frac{1}{\sqrt{6}}(2.5 \times 1 + (-2.5) \times (-1) + 5 \times 2) = \frac{15}{\sqrt{6}}.$$

So,

$$\text{proj}_{\mathbf{q}_1} \mathbf{v}_3 = \frac{15}{\sqrt{6}}\mathbf{q}_1 = \mathbf{v}_3.$$

Since $\text{proj}_{\mathbf{q}_2} \mathbf{v}_3 = 0$, we have:

$$\tilde{\mathbf{q}}_3 = \mathbf{v}_3 - \text{proj}_{\mathbf{q}_1} \mathbf{v}_3 - \text{proj}_{\mathbf{q}_2} \mathbf{v}_3 = \mathbf{v}_3 - \mathbf{v}_3 - \mathbf{0} = \mathbf{0}.$$

$\tilde{\mathbf{q}}_3$ is zero, so we cannot define $\mathbf{q}_3$.

Since we obtained two orthonormal vectors, the subspace spanned by $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ has a dimension of **2**.

Problem 5. (**6 pts**) Consider a dataset with many features, some of which are irrelevant or redundant.

(a) Describe the difference between Lasso and Ridge regression in terms of their effect on the coefficients of the model.

(b) Formulate the loss function for Lasso regression, and explain why this optimization problem leads to sparse solutions.

(c) Discuss how the choice of the regularization parameter λ affects the sparsity of the solution.

Solution:

(a) - Lasso Regression (L1 Regularization): Uses the $L1$ norm $\|\theta\|_1$ in the penalty term. It can shrink some coefficients to exactly zero, resulting in a sparse model that selects a subset of features.

- Ridge Regression (L2 Regularization): Uses the $L2$ norm $\|\theta\|_2^2$ in the penalty term. It shrinks coefficients towards zero but does not set them exactly to zero, retaining all features but with reduced impact.

(b)

$$L(\theta) = \frac{1}{2}\|A\theta - y\|^2 + \lambda\|\theta\|_1,$$

The $L1$ norm penalty $\lambda\|\theta\|_1$ introduces a non-differentiable point at zero for each coefficient. This creates a sharp corner ("kink") in the cost function at zero, making it more likely for the optimization process to set coefficients exactly to zero to minimize the overall cost. This leads to sparse solutions with many zero coefficients.

(c) - Increasing λ: Strengthens the penalty on the magnitude of the coefficients, promoting greater sparsity by driving more coefficients to zero.

- Decreasing λ: Weakens the penalty, allowing more coefficients to remain non-zero, thus including more features in the model.

The choice of λ balances model complexity and sparsity, affecting the number of features selected.

Problem 6. (**4 pts**) Suppose a_1, a_2 is a list of two linearly independent n-vectors. When we run the Gram-Schmidt algorithm on this list, we obtain the orthonormal vectors q_1, q_2. Now suppose we run the Gram-Schmidt algorithm on the list of vectors a_2, a_1 (i.e., the same vectors, in reverse order). Do we get the orthonormal vectors q_2, q_1 (i.e., the orthonormal vectors obtained from the original list, in reverse order)?

If you believe this is true, give a very brief explanation why. If you believe it is not true, give a simple counter-example. Hint: Use geometrical interpretation of Gram-Schmidt process.

Solution:

The statement is **not true**. We do not get q_2, q_1 if we reverse the order of the vectors and apply the Gram-Schmidt process.

The Gram-Schmidt process depends on the order of the vectors. When running the Gram-Schmidt process on a_1, a_2, the first orthonormal vector q_1 is obtained by normalizing a_1. The second orthonormal vector q_2 is then obtained by removing the component of a_2 along q_1 and normalizing the resulting vector. This process produces vectors that are orthogonal to each other.

However, if we reverse the order and run Gram-Schmidt on a_2, a_1:

- The first orthonormal vector q_1' is now obtained by normalizing a_2.
- The second orthonormal vector q_2' is then obtained by removing the component of a_1 along q_1' and normalizing the result.

Since a_1 and a_2 are generally not aligned in the same direction, the projections and resulting orthogonal components will differ depending on the order. Thus, q_2, q_1 are generally not the same as q_1', q_2', because the vector directions and magnitudes of the intermediate steps depend on the order in which vectors are processed. You can also illustrate this with the help of a simple example.

1. **Order a_1, a_2:**

$$q_1 = \frac{a_1}{\|a_1\|} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

$$q_2 = \frac{a_2 - (q_1^T a_2) q_1}{\|a_2 - (q_1^T a_2) q_1\|} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

2. **Order a_2, a_1:**

$$q_1' = \frac{a_2}{\|a_2\|} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

$$q_2' = \frac{a_1 - (q_1'^T a_1) q_1'}{\|a_1 - (q_1'^T a_1) q_1'\|} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

In this example, $q_2 \neq q_1'$ and $q_1 \neq q_2'$, showing that reversing the order changes the resulting orthonormal vectors.

Problem 7. (**4 pts**)

Suppose $q_1, \ldots, q_k$ is an orthonormal set of n-vectors, with $k < n$. We wish to determine if a given n-vector a is a linear combination of them.

Your nearest neighbor (in the exam hall) has a claim that this is easy to do. He/she says that we compare the n-vector

$$\hat{a} = (q_1^T a)q_1 + \cdots + (q_k^T a)q_k$$

to a. If $\hat{a} = a$, then a is a linear combination of $q_1, \ldots, q_k$. If $\hat{a} \neq a$, then a is not a linear combination of $q_1, \ldots, q_k$.

Obviously, you cannot approach your neighbor to verify the claim at this point in time.

Do you agree with your neighbor? Give your answer as yes or no, and add a very short explanation.

Solution:

Yes, your neighbor is correct.

If a is a linear combination of the orthonormal vectors $q_1, \ldots, q_k$, then it can be written as:

$$a = c_1 q_1 + c_2 q_2 + \cdots + c_k q_k,$$

for some scalars $c_1, \ldots, c_k$. In this case, the projection of a onto the span of $q_1, \ldots, q_k$ is exactly a, so $\hat{a} = a$.

If a is not in the span of $q_1, \ldots, q_k$, then the projection $\hat{a}$ only captures the component of a along the span of these vectors, and $\hat{a} \neq a$. Thus, checking whether $\hat{a} = a$ correctly determines whether a is a linear combination of the given orthonormal set.

Problem 8. (**6 pts**) Given a vector $\mathbf{v} = (2, -1, 3)$ and a subspace spanned by the vectors:

$$\mathbf{u}_1 = (1, 2, 0), \quad \mathbf{u}_2 = (0, 1, 1),$$

find the **projection of v** onto the subspace spanned by $\mathbf{u}_1$ and $\mathbf{u}_2$.

Solution:

The projection of $\mathbf{v}$ onto a vector $\mathbf{u}_i$ is given by:

$$\text{proj}_{\mathbf{u}_i}\mathbf{v} = \frac{\mathbf{v} \cdot \mathbf{u}_i}{\mathbf{u}_i \cdot \mathbf{u}_i}\mathbf{u}_i$$

We need to find the sum of the projections onto both $\mathbf{u}_1$ and $\mathbf{u}_2$:

$$\text{proj}_{\text{span}(\mathbf{u}_1,\mathbf{u}_2)}\mathbf{v} = \text{proj}_{\mathbf{u}_1}\mathbf{v} + \text{proj}_{\mathbf{u}_2}\mathbf{v}.$$

Projection onto $\mathbf{u}_1$:

$$\text{proj}_{\mathbf{u}_1}\mathbf{v} = \frac{0}{5}(1, 2, 0) = (0, 0, 0).$$

Projection onto $\mathbf{u}_2$:

$$\text{proj}_{\mathbf{u}_2}\mathbf{v} = (0, 1, 1).$$

Final Projection: The projection of $\mathbf{v}$ onto the subspace spanned by $\mathbf{u}_1$ and $\mathbf{u}_2$ is:

$$\text{proj}_{\text{span}(\mathbf{u}_1,\mathbf{u}_2)}\mathbf{v} = (0, 0, 0) + (0, 1, 1) = (0, 1, 1).$$

Problem 9. (**10 pts**)

You are given a dataset with five observations and four predictors:

Observation	x_1	x_2	x_3	x_4	y
1	1	2	3	4	20
2	2	4	6	8	40
3	3	6	9	12	60
4	4	8	12	16	80
5	5	10	15	20	100

Please note the perfect multicollinearity among the predictors: $x_2 = 2x_1$, $x_3 = 3x_1$, $x_4 = 4x_1$.

(a) Attempt to perform a standard linear regression of y on x_1, x_2, x_3, and x_4 without regularization. Explain why the Moore-Penrose pseudo-inverse cannot be used directly in this case.

(b) Remove the redundant features and formulate a model relating the input(s) and output. Use pseudo-inverse to determine the parameters of the model.

(a) Attempt to perform a standard linear regression of y on x_1, x_2, x_3, and x_4 without regularization.

First, construct the design matrix X (including an intercept term) and response vector y:

$$X = \begin{bmatrix} 1 & 1 & 2 & 3 & 4 \\ 1 & 2 & 4 & 6 & 8 \\ 1 & 3 & 6 & 9 & 12 \\ 1 & 4 & 8 & 12 & 16 \\ 1 & 5 & 10 & 15 & 20 \end{bmatrix}, \quad y = \begin{bmatrix} 20 \\ 40 \\ 60 \\ 80 \\ 100 \end{bmatrix}$$

Let $\boldsymbol{\beta} = [\beta_0, \beta_1, \beta_2, \beta_3, \beta_4]^\top$.

The matrix $X^\top X$ is **singular** (non-invertible) due to perfect multicollinearity among the predictors. Specifically, since x_2, x_3, and x_4 are exact linear multiples of x_1, the columns of X are linearly dependent. This results in $\det(X^\top X) = 0$, making it impossible to compute $(X^\top X)^{-1}$ required to solve for estimating parameters.

(b) Compute the regression coefficients using the Moore-Penrose pseudoinverse.

We can reduce the model by expressing all predictors in terms of x_1:

$$x_2 = 2x_1, \quad x_3 = 3x_1, \quad x_4 = 4x_1$$

Therefore, the model simplifies to:

$$y = \beta_0 + (\beta_1 + 2\beta_2 + 3\beta_3 + 4\beta_4)x_1$$

Let $\tilde{\beta} = \beta_1 + 2\beta_2 + 3\beta_3 + 4\beta_4$.

Construct the reduced design matrix X_{reduced}:

$$X_{\text{reduced}} = \begin{bmatrix} 1 & x_1 \\ 1 & x_1 \\ 1 & x_1 \\ 1 & x_1 \\ 1 & x_1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{bmatrix}$$

Compute $X_{\text{reduced}}^\top X_{\text{reduced}}$:

$$X_{\text{reduced}}^\top X_{\text{reduced}} = \begin{bmatrix} 5 & 15 \\ 15 & 55 \end{bmatrix}$$

Compute its inverse:

$$(X_{\text{reduced}}^\top X_{\text{reduced}})^{-1} = \frac{1}{50} \begin{bmatrix} 55 & -15 \\ -15 & 5 \end{bmatrix}$$

(ii) Compute $\boldsymbol{\beta} = (X^\top X)^+ X^\top y$.

Compute $X_{\text{reduced}}^\top y$:

$$X_{\text{reduced}}^\top y = \begin{bmatrix} 300 \\ 1100 \end{bmatrix}$$

Compute the coefficients:

$$\begin{bmatrix} \beta_0 \\ \tilde{\beta} \end{bmatrix} = (X_{\text{reduced}}^\top X_{\text{reduced}})^{-1} X_{\text{reduced}}^\top y = \frac{1}{50} \begin{bmatrix} 55 & -15 \\ -15 & 5 \end{bmatrix} \begin{bmatrix} 300 \\ 1100 \end{bmatrix} = \begin{bmatrix} 0 \\ 20 \end{bmatrix}$$

Therefore, $\beta_0 = 0$ and $\tilde{\beta} = 20$. Since $\tilde{\beta} = \beta_1 + 2\beta_2 + 3\beta_3 + 4\beta_4$, we cannot uniquely determine individual β_j coefficients due to multicollinearity. The Moore-Penrose pseudoinverse provides the solution with the smallest Euclidean norm, which, in this context, implies distributing the coefficient $\tilde{\beta}$ in a way that minimizes $\|\boldsymbol{\beta}\|_2$.

Problem 10. (**16 pts**)

You have been invited on an extraordinary journey into the heart of a hidden melody encoded within a mysterious matrix. Your task is to decode the secret patterns that bring harmony to the Symphony of Sound. Solving these challenges will reveal the composition and unlock the grand finale of the symphony — the elusive Resonant Note!

You are given an $m \times n$ matrix A, containing information about musical notes. Below are five musical clues that will guide your way:

1. **Harmonic Balance:** The inner matrix A^TA has two special eigenvalues, 25 and 9, representing two dominant frequencies in the composition.
2. **Chord Vector:** One of the eigenvectors of A^TA is $w_1 = \frac{1}{\sqrt{10}}[3 \quad 1]^T$, which serves as a key interval in the harmonic structure.
3. **Note Patterns:** The left singular matrix U holds the essential patterns of the melody. One crucial part of the melody is encoded in the first column, $u_1 = \frac{1}{\sqrt{2}}[1 \quad 1 \quad 0]^T$.
4. **The Melody's Journey:** Two musical phrases, p_1 and p_2, play important roles. They create the following transformations when played through the matrix A:

$$Ap_1 = \frac{5}{\sqrt{2}}[1 \quad 1 \quad 0]^T, \quad Ap_2 = [0 \quad 0 \quad 3]^T.$$

   These phrases correspond to the first two columns of the right singular matrix V.
5. **Resonant Note:** At the heart of this composition lies the singular value matrix Σ. The note that concludes the symphony corresponds to the value of Σ_{22}.

Decode the clues and unlock the final note of the symphony as follows:

(a) (2 marks) **Reveal the Composition:** Determine the dimensions of the musical matrix A.

(b) (5 marks) **Uncover the Patterns:** Compute the left singular matrix, U, to reveal the underlying melody.

(c) (5 marks) **Chart the Musical Phrases:** Find the right singular matrix, V, to understand the structure of the musical phrases.

(d) (3 marks) **Discover the Resonance:** Calculate the singular values in Σ to understand the harmonic structure.

(e) (1 mark) **Find the Final Note:** Determine the value of Σ_{22}, the Resonant Note of the symphony.

Solution:

We are given that A has SVD $A = U\Sigma V^T$ with

$$A^TA \text{ has eigenvalues } 25,\ 9, \quad w_1 = \frac{1}{\sqrt{10}}\begin{bmatrix}3\\1\end{bmatrix}, \quad u_1 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\\0\end{bmatrix}, \quad Ap_1 = \frac{5}{\sqrt{2}}\begin{bmatrix}1\\1\\0\end{bmatrix},\ Ap_2 = \begin{bmatrix}0\\0\\3\end{bmatrix}.$$

(a) *Dimensions of A.* Since $u_1 \in \mathbb{R}^3$ and $w_1 \in \mathbb{R}^2$, we have

$$A \in \mathbb{R}^{3\times 2} \quad (m = 3,\ n = 2).$$

(b) *Left singular matrix U.* From clue 3: $u_1 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\\0\end{bmatrix}$

From clue 4: $Ap_2 = \begin{bmatrix}0\\0\\3\end{bmatrix}$, and since $p_2 = v_2$ and $Av_2 = \sigma_2 u_2$, with $\sigma_2 = 3$ (from eigenvalues 25 and 9 giving singular values 5 and 3), we have:

$$3u_2 = \begin{bmatrix}0\\0\\3\end{bmatrix} \Rightarrow u_2 = \begin{bmatrix}0\\0\\1\end{bmatrix}$$

We need u_3 orthogonal to both u_1 and u_2, and unit length:

$$u_3 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\\0\end{bmatrix}$$

Thus:

$$U = \begin{bmatrix}\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}}\\ 0 & 1 & 0\end{bmatrix}$$

(c) *Right singular matrix V.* The right singular vectors are eigenvectors of A^TA. We are given $w_1 = \frac{1}{\sqrt{10}}[3,1]^T$, which we take as v_1 (the eigenvector for eigenvalue 25). A unit vector orthogonal to v_1 is

$$v_2 = \frac{1}{\sqrt{10}}\begin{bmatrix}-1\\3\end{bmatrix} \quad \big([3,1]\cdot[-1,3] = 0,\ \|v_2\| = 1\big).$$

Hence

$$V = \begin{bmatrix}\frac{3}{\sqrt{10}} & -\frac{1}{\sqrt{10}}\\ \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}}\end{bmatrix}.$$

(d) *Singular values* Σ. From the eigenvalues 25 and 9 of A^TA, the singular values are

$$\sigma_1 = 5, \qquad \sigma_2 = 3.$$

With the compact SVD for $A \in \mathbb{R}^{3\times 2}$,

$$\Sigma = \begin{bmatrix}5 & 0\\ 0 & 3\\ 0 & 0\end{bmatrix}.$$

(e) *Final note* Σ_{22}.

$$\boxed{\Sigma_{22} = 3.}$$